Charlotte Cornell

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EDUCATION

Cognitive Psychology (Ph.D. Candidate, M.S.) Sept. 2023 – present
Rutgers University – New Brunswick, NJ, USA
Mathematics & Psychology (B.S.), *honors, magna cum laude* May 2021
Loyola University Chicago – Chicago, IL, USA

HONORS & AWARDS

- Institute of Teaching, Innovation, and Inclusive Pedagogy of Rutgers University's "You've Been Thanked" Teaching Certificate — Jan. 2025
- Women of Mathematical Psychology's Professional Representation for Inclusivity and Minority Empowerment Travel and Networking Award — July 2024
- Mathematics Departmental Honors — May 2021
- Psychology Departmental Honors — May 2021
- Loyola University Chicago's Interdisciplinary Honors Program — Aug. 2017 – May 2021

RELEVANT EXPERIENCE

Graduate Student / Teaching Assistant, Sept. 2023 – present
& Lab Manager / Research Assistant Sept. 2021 – Aug. 2023
Psychology Department, Rutgers University–New Brunswick, USA

- Design computational models of memory search for naturalistic stimuli by reviewing literature, designing and analyzing behavioral experiments, and writing and presenting results.
- Mentor undergraduate students on research projects bridging psychological and computer sciences.
- Manage research participant information, lab accounts, expenses, and protocols.
- Teach undergraduate lab courses on conducting and writing about experiments of human cognition.

Research Assistant Jan. 2020 – Dec. 2022
Psychology Department, Loyola University Chicago, USA

- Developed a novel model analysis method by building and testing models of group interaction, analyzing simulations, and writing results.
- Researched sleep health and memory by reviewing literature, designing and analyzing experiments, and writing and presenting results.

Teaching Assistant Jan. 2019 – May 2019
Computer Science Department, Loyola University Chicago, USA

- Taught data analysis languages by instructing code tutorials and providing assignment feedback.
- Assisted students with projects by helping them develop goals and data visualizations.

PUBLICATIONS

* shared first-authorship

Journal Articles

Cornell, C.A., & Zhang, Q. (under review). Hierarchical organization of context guides efficient memory search. [link]

Angne, H., **Cornell, C.A.**, & Zhang, Q. (2024). A context-based model of collaborative inhibition during memory search. *Scientific Reports, 14,* Article 27645. [link]

Cornell, C.A., Norman, K.A., Griffiths, T.L., & Zhang, Q. (2024). Improving memory search through model-based cue selection. *Psychological Science, 35(1),* 55-71. [link]

Larson, J.R., Jr., **Cornell, C.A.,** & Aramovich, N.P. (2024). Building better theories: Prediction intervals as a tool for theory testing and improvement. *Basic and Applied Social Psychology, 46(3),* 146-168. [link]

Conference Proceedings

Cornell, C.A.*, Jin, S.*, & Zhang, Q. (2024). The role of episodic memory in storytelling: Comparing large language models with humans. *Proceedings of the 46th Annual Conference of the Cognitive Science Society.* [link]

Angne, H., **Cornell, C.A.**, & Zhang, Q. (2024). Why two heads together are worse than apart: A context-based account of collaborative inhibition in memory. *Proceedings of the 46th Annual Conference of the Cognitive Science Society.* [link]

Book Chapters

Yoon, Y. & **Cornell, C.A.** (in press). Indirect social influence via shared environments. In J. R. Larson & C. Coen (Eds.), *Agent-Based Modeling for Research on Groups, Networks, and Organizations*. American Psychological Association.

PRESENTATIONS

*presenting author

Cornell, C.A.* & Zhang, Q. (Nov. 2026). *A computational model of how external cues facilitate memory of events.* Poster presentation to be given at the 67th Annual Meeting of the Psychonomics Society, San Diego, CA, USA.

Cornell, C.A.* & Zhang, Q. (July 2026). *A computational model of how external cues facilitate memory of events.* Spoken presentation given at the joint 59th Annual Meeting of the Society for Mathematical Psychology and 24th International Conference on Cognitive Modeling, Montréal, Québec, Canada.

Cornell, C.A.* & Zhang, Q. (May 2026). *Hierarchical context guides efficient memory search.* Poster presentation given at the 21st Annual Context and Episodic Memory Symposium, Philadelphia, PA, USA.

Cornell, C.A.* & Zhang, Q. (May 2026). *Hierarchical context guides efficient memory search.* Spoken presentation given at Psychology Department's Cog Fest at Rutgers University, Piscataway, NJ, USA.

Cornell, C.A.* & Zhang, Q. (Nov. 2025). *A hierarchical model of within-list and across-list recall.* Spoken presentation given at the 66th Annual Meeting of the Psychonomics Society, Denver, CO, USA.

Cornell, C.A.* & Zhang, Q. (July 2025). *A unified model of within-list and across-list memory search.* Spoken presentation given at the joint 58th Annual Meeting of the Society for Mathematical Psychology and 23rd International Conference on Cognitive Modeling, Ohio State University, Columbus, Ohio, USA.

Cornell, C.A.* & Zhang, Q. (April 2025). *Improving memory search: A context-based account*. Spoken presentation given at Psychology Department's Cog Fest at Rutgers University, Piscataway, NJ, USA.

Cornell, C.A.* & Zhang, Q. (April 2025). *Modeling cued memory search*. Spoken presentation given at the Psychology Department's Data Blitz at Rutgers University, Piscataway, NJ, USA.

Cornell, C.A.*¸ Jin, S., & Zhang, Q. (Nov. 2024). *The role of episodic memory in storytelling: Comparing large language models with humans.* Poster presentation given at the Satellite Meeting of the Society for Mathematical Psychology at the 65th Meeting of the Psychonomic Society, New York City, New York.

Cornell, C.A.*¸ Jin, S., & Zhang, Q. (July 2024). *The role of episodic memory in storytelling: Comparing large language models with humans.* Poster presentation given at the 46th Annual Conference of the Cognitive Science Society, Rotterdam, the Netherlands.

Angne, H.*, **Cornell, C.A.**, & Zhang, Q. (July 2024). *Why two heads together are worse than apart: A context-based account of collaborative inhibition in memory.* Poster presentation given at the 46th Annual Conference of the Cognitive Science Society, Rotterdam, the Netherlands.

Angne, H.*, **Cornell, C.A.**, & Zhang, Q. (July 2024). *Why two heads together are worse than apart: A context-based account of collaborative inhibition in memory.* Spoken presentation given at the joint 57th Annual Meeting of the Society for Mathematical Psychology and 22nd International Conference on Cognitive Modeling, Tilburg University, the Netherlands.

Cornell, C.A.*, Jin, S., & Zhang, Q. (July 2024). *The role of episodic memory in storytelling: Comparing large language models with humans.* Spoken presentation given at the joint 57th Annual Meeting of the

Society for Mathematical Psychology and 22nd International Conference on Cognitive Modeling, Tilburg University, the Netherlands.

Cornell, C.A.*, Norman, K.A., Griffiths, T.L., & Zhang, Q. (May 2024). *Improving memory search through model-based cue selection*. Poster presentation given at the 20th Annual Context and Episodic Memory Symposium, Philadelphia, PA, USA.

Cornell, C.A.*, Norman, K.A., Griffiths, T.L., & Zhang, Q. (May 2024). *Improving memory search through model-based cue selection*. Spoken presentation given at the Psychology Department's Cognitive Fest at Rutgers University, Piscataway, NJ, USA.

Cornell, C.A.*, Norman, K.A., Griffiths, T.L., & Zhang, Q. (Nov. 2023). *Improving memory search through model-based cue selection*. Spoken presentation given at the 64th Annual Meeting of the Psychonomics Society, San Francisco, CA, USA.

Cornell, C.A.*, Norman, K.A., Griffiths, T.L., & Zhang, Q. (July 2023). *Improving memory search through model-based cue selection*. Spoken presentation given at the joint 56th Annual Meeting of the Society for Mathematical Psychology and 21st International Conference on Cognitive Modeling at the University of Amsterdam, the Netherlands.

Cornell, C.A.*, & Leach, R.C. (May 2021). *Fatigue, drowsiness, and emotional working memory*. Spoken presentation given at the Undergraduate Research & Engagement Symposium at Loyola University Chicago, Chicago, IL, USA.

PROFESSIONAL AFFILIATIONS

2024 – Member, Cognitive Science Society
2023 – Member, Psychonomic Society
2023 – Member, Society for Mathematical Psychology
2023 – Member, Women of Mathematical Psychology

REFERENCES

- **Qiong Zhang, Ph.D.**
  Graduate Research Advisor
  Assistant Professor
  Psychology & Computer Science Departments
  Rutgers University
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- **Jim Larson, Ph.D.**
  Undergraduate Research Advisor
  Professor Emeritus
  Psychology Department
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